

System startup sequence

ESP32 host

1. app_main

chip and persistent epoch

2. Coordinator runtime

parser, sessions, routes

3. Network services

LAN, mDNS, Host RPC, SNTP

4. User services

WebShell, Lua, inference

STM32 node

1. HAL + peripherals

UART DMA, timers, display

2. FreeRTOS objects

queues, pools, static tasks

3. Scheduler starts

link, mesh, service, compute

4. Link probing

HELLO / ACK on every UART

bootstrap traffic begins

Both sides become observable before application jobs are accepted.